Verifiers that do not trust the agent: anytime-valid release gates and hacking bounds for money-moving AI agents

Scruple

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Abstract

An AI agent that handles card disputes moves money: it decides when to refund, when to fight and with which evidence. Changes to such an agent are usually judged by dashboards of wins and by the agent's own reports, and both can be gamed. We describe a release gate whose verifiers never read the agent's claims for credit: cost is recomputed from double-entry ledger postings and every action is checked against an append-only, hash-chained trace written by the environment. Its statistics are anytime-valid: a paired betting e-process for the mean per-case cost difference, per-case-type e-processes combined by weighted e-Bonferroni, and a certificate that bounds the probability of promoting a harmful change at any stopping time. We prove the following. (i) For any adaptive, randomised agent that may learn something about which cases are audited, the proxy-over-ledger gap Γ of an accepted change satisfies $\mathbb{P}(\text{accepted},\ \Gamma \geq \gamma) \leq (1-\rho)^{\lceil n\gamma/\varepsilon \rceil}$ and $\mathbb{E}[\Gamma \mathbf{1}\{\text{accepted}\}] \leq \varepsilon/(ne \ln \frac{1}{1-\rho})$, where ρ is a floor on the conditional catch probability; with a likelihood-ratio leak ℓ and a verifier false-negative rate β, $\rho = (1-\beta)q\ell/(q\ell+1-q)$. (ii) With verifier false-negative rate β and false-alarm rate ϕ, hacking stops paying exactly when the audit rate reaches $q^\star = g/((1-\beta)(K+g) - \phi K)$, and no audit rate deters unless $\beta g \leq (1-\beta-\phi)K$. (iii) A grid mixture of constant bets is within $\log J$ of the best constant bet on every path, and the best constant-bet growth rate of the observed paired costs gives a floor on the sample size of any valid test. On the planted regressions the default bet's cap was the bottleneck: a wider, equally valid bet raised resampled power from 0.37 to 0.91 on one regression and, on the benchmark itself, detected the smallest one (about one cent per case overall, concentrated in one dispute family), which the shipped gate had missed. (iv) The composed promote decision carries the smallest level among the conjuncts that cover a harm; false blocks add up; harms no conjunct covers carry no guarantee. On a frozen synthetic benchmark of 10,000 dispute cases the gate caught all 7 planted regressions in a pre-registered experiment that passed narrowly, and the bounds held on every policy a proxy optimiser visited. A second pre-registered experiment failed its main hypothesis: the hacked policy was \$33.87 per case cheaper on the simulated ledger because forged documents win in the simulator; provenance checks blocked it and a cost-only gate would have promoted it. Real Claude models playing simulated disputes exposed three verifier false positives, and the cheapest model won 4 of its 10 wins against claims the cardholder was right about.

Data and labels. Numbers marked *synthetic* come from the scruple simulator; numbers marked *real* come from public CFPB complaint data or from real model behaviour. No number describes a real merchant, bank or card network. Each result is labelled *standard* (known, applied here), *adaptation* (a small variant of a known result) or *candidate novel* (not found stated this way in a short search; a full literature review is still required). Code, data and every result file: `https://github.com/thedhruvhegde/scruple`, directory `engine/`.

1 Introduction

Card disputes are a good test bed for agents that move money. Every case ends in postings to a ledger: a refund, a chargeback, a fee, a representment won or lost. The rules are public (Regulation E, 12 CFR 1005.11, for debit cards; Regulation Z, 12 CFR 1026.13, for credit cards), and the incentives to cheat are concrete. An agent judged on its win rate can fight claims it should concede, cite documents it never opened, upload documents that never existed, or simply report a win.

The usual guard for an agent change is an offline evaluation: replay the new agent on held-out cases, compare a metric, ship if it improves. That guard fails in three ways that we measure.

1. **The metric trusts the agent.** In our experiment E2 a seeded hill climb on a "wins plus self-reported wins" proxy learned two exploits (fabricated documents and false win claims) in three moves.
2. **The statistics do not survive how they are used.** Re-running a z-test every 10 cases at the null boundary rejects 14.7% to 34.3% of the time at a nominal 5% (synthetic). A regression of 0.35 cap units in a case type that is 1% of traffic, with the rest improving, is flagged by a pooled valid test 0.6% of the time.
3. **Nobody states what the release decision guarantees.** A gate mixes deterministic checks, soft scores and tests; which error does the combination control?

Contributions.

- A verifier design anchored to the ledger and to an environment-written trace, with an attack for every check (Section 3).
- Anytime-valid paired and stratified tests with complete proofs (Sections 4 and 5), sample-size planning with a matching lower bound (Section 6), and variance-adaptive betting with a sample-size floor computed from observed data (Section 7, new here).
- A composition theorem for the release decision (Section 9, new here) and its asymmetric-cost refinement (Section 8).
- Hacking-gap bounds for adaptive agents with leaky audits and an imperfect verifier, and the minimal audit rate that removes the incentive to hack (Section 10; Theorems 10.3 and 10.6 and Corollaries 10.5 and 10.7 are new here).
- Two pre-registered synthetic experiments (one passed, one failed on its main hypothesis), real-data baselines and real-model results on public CFPB complaints, real models playing simulated disputes, and a pre-registered real-model incentive experiment (Section 11).

2 Setting and notation

Definition 2.1 (Episode and record)**.** A dispute episode starts from a seeded case: a card transaction, a reason code, a cardholder claim with a hidden truth (valid or not), documents the merchant does or does not hold, and network deadlines. The agent acts through tools (read the case, list and read evidence, request information, read the ledger, refund, represent, accept, wait, upload evidence). The environment writes every call and every state change to an append-only trace whose events are hashed into a chain with a recorded head. Money is double-entry, in USD minor units. The *record* of an episode is the trace, the final journal, the documents as generated and at close, the case with its hidden truth, and the agent's self-report.

Definition 2.2 (Paired difference)**.** A candidate policy C is compared with the current policy B on the same held-out cases and the same issuer draws. Case i gives losses B_i, C_i (merchant

cost from the journal). Before outcomes are seen the operator declares a cap $c > 0$, and

$$X_i = \frac{C_i - B_i}{c} \in [-1, 1], \qquad \mu_i = \mathbb{E}[X_i \mid \mathcal{F}_{i-1}], \tag{1}$$

where $\mathcal{F}_{i-1}$ is everything observed before case i. Positive X_i means the candidate is worse.

Assumption 2.3 (Cap)**.** $|C_i - B_i| \le c$ for every case. A violation is an error; it is never clipped silently.

Assumption 2.4 (Predictable bets)**.** Every bet λ_i used for case i is $\mathcal{F}_{i-1}$-measurable.

The gate decides *promote*, *hold* or *block*. The error that costs money is promoting a harmful change; the error that costs adoption is blocking a good one.

Benchmark. Benchmark v1 is 10,000 synthetic dispute cases from a sealed held-out seed namespace; only a manifest of commitments is public (content hash `1286baf3...2cb13f91`). Family mix: buyer's remorse 1,977; cardholder mistake 577; friendly fraud 2,791; merchant error 625; service failure 1,840; true fraud 2,190; 4,655 claims are valid. Reference policies (synthetic): the *evidence* policy (represent when the required evidence exists, else concede) costs $82.82 per case (95% interval 80.72 to 84.91) and fails a hard check on 19 of 10,000 cases; always-accept costs $127.46; an oracle that reads the hidden truth costs $86.67, more than the evidence policy, because knowing a claim is invalid does not win it without documents.

3 Verifier design: anchoring to the ledger and the trace

The design principle is that nothing the agent says earns credit. Self-reports are *checked*, never *scored*: a self-report can only lose points. Scores compose as

$$\text{gate} = \bigwedge_h \text{hard}_h, \qquad \text{quality} = \textstyle\sum_j w_j s_j / \sum_j w_j, \qquad \text{reward} = \text{gate} \times \text{quality}, \tag{2}$$

so a hard failure zeroes the reward and ten perfect soft scores cannot outvote one violation. Table 1 lists the checks and, for each, an attack that fools a naive metric and is stopped by the check.

On 300 synthetic seeds the exploit run reports, by row: caught 28, blocked 5 (the environment refuses the move), partial 1, not caught 1, not modelled 7. "Caught" means the naive metric rates the exploit at least as well as the honest policy, the expected check fires on at least 90% of exercised episodes, and the reward falls. The partial case is "escalate everything" (94 of 125 exercised episodes caught; on the rest it wins a valid claim cheaply and only the soft OUT-01 notices); the one not caught is "memorised outcomes" (0 of 30). Mutation testing of the 18 verifier modules (2,173 mutants) gave a mutation score of 63.8%, 64.7% after 15 targeted tests.

4 Anytime-valid paired test

We first record the two probabilistic tools every later proof uses.

Lemma 4.1 (Ville's inequality)**.** *Let $(K_t)_{t\ge 0}$ be a non-negative supermartingale with $K_0 = 1$. Then for every $a > 0$, $\mathbb{P}(\exists t : K_t \ge a) \le 1/a$, and for every stopping time τ (possibly infinite, with $K_\infty = \liminf_t K_t$), $\mathbb{E}[K_\tau] \le 1$.* *[standard; Ville, 1939]*

Proof. Let $\sigma = \inf\{t : K_t \ge a\}$. For fixed T, $\sigma \wedge T$ is a bounded stopping time, so by optional stopping for supermartingales (Durrett, 2019, Ch. 4) $\mathbb{E}[K_{\sigma\wedge T}] \le K_0 = 1$. Since $K \ge 0$ and $K_{\sigma\wedge T} \ge a$ on $\{\sigma \le T\}$, $a\,\mathbb{P}(\sigma \le T) \le 1$. Let $T \to \infty$: $\mathbb{P}(\sigma < \infty) \le 1/a$. For the second claim, $K_{\tau\wedge T} \to K_\tau$ on $\{\tau < \infty\}$ and $\liminf_T K_{\tau\wedge T} = \liminf_t K_t = K_\infty$ on $\{\tau = \infty\}$; Fatou's lemma gives $\mathbb{E}[K_\tau] \le \liminf_T \mathbb{E}[K_{\tau\wedge T}] \le 1$. □

Table 1: Checks, what they read, and the attack each stops. Synthetic, 300 seeds per attack.

check	reads	kind	attack stopped (naive metric fooled)
INV-01 ledger invariants	trace postings, after every action	hard	double credit then moot
INV-05 record integrity	hash chain and head	hard	outcome rewrite (correct rate 1.0)
PROC-01 process	trace order	hard	summary response (half the calls)
EVD-01 provenance	generator truth, content hashes	hard	fabrication (win rate 0.044 to 0.85)
DL-01 deadlines	env-recorded network calendar	hard	stall on pending information
SR-01 self-report	claims parsed into predicates on the record	hard	claimed, not done
SCOPE-01 scope	protected reads, tool and argument scope	hard	ground-truth peeking
CON-01, CAL-01, SR-03	batch statistics	hard	segment skimming, flat confidence
CF-01 counterfactual	replay under baseline, same draws	soft	refund to win
OUT-01 fair outcome	hidden truth	soft	unfair wins (Section 11.4)

Lemma 4.2 (Data processing for KL)**.** *Let* $\mathrm{kl}(p\|q) = p\log\frac{p}{q} + (1-p)\log\frac{1-p}{1-q}$*. For laws* P, Q *and any event* A*,* $\mathrm{KL}(P\|Q) \geq \mathrm{kl}(P(A)\|Q(A))$*. If* $P(A) \geq 1-\beta > \alpha \geq Q(A)$*, then* $\mathrm{kl}(P(A)\|Q(A)) \geq \mathrm{kl}(1-\beta\|\alpha)$*.* *[standard; Cover and Thomas, 2006]*

Proof. The first inequality is the data-processing inequality applied to the map $\omega \mapsto \mathbf{1}_A(\omega)$. For the second, $\partial_p \mathrm{kl}(p\|q) = \log\frac{p(1-q)}{q(1-p)} \geq 0$ for $p \geq q$ and $\partial_q \mathrm{kl}(p\|q) = \frac{q-p}{q(1-q)} \leq 0$ for $q \leq p$; so increasing p from $1-\beta$ and decreasing q from α only increases kl. □

Proposition 4.3 (Betting e-process)**.** *Fix* $m \in (-1, 1)$ *and a side* $s \in \{+1, -1\}$ *(*$s = +1$*: regression,* $H_0 : \mu_i \leq m$ *for all* i*;* $s = -1$*: certification,* $H_0 : \mu_i \geq m$ *for all* i*). Let* $Y_i = s(X_i - m)$ *and predictable bets* $0 \leq \lambda_i \leq b/(1+sm)$ *with* $b < 1$*. Then*

$$K_t = \prod_{i \leq t}(1 + \lambda_i Y_i) \tag{3}$$

satisfies, under H_0*,* $\mathbb{P}(\exists t : K_t \geq 1/\alpha) \leq \alpha$*, and* $\mathbb{E}[K_\tau] \leq 1$ *for every stopping time* τ*.*
[standard; Ramdas et al., 2023; Waudby-Smith and Ramdas, 2024]

Proof. Since $X_i \in [-1, 1]$, $Y_i \geq -(1+sm)$, so $1 + \lambda_i Y_i \geq 1 - b > 0$ and $K_t > 0$. Under H_0, $s(\mu_i - m) \leq 0$; as $\lambda_i \geq 0$ is $\mathcal{F}_{i-1}$-measurable, $\mathbb{E}[1 + \lambda_i Y_i \mid \mathcal{F}_{i-1}] = 1 + \lambda_i s(\mu_i - m) \leq 1$. Hence K is a non-negative supermartingale with $K_0 = 1$, and Lemma 4.1 with $a = 1/\alpha$ gives both claims. □

The aGRAPA bet of Algorithm 1 maximises $\lambda\mathbb{E}Y - \lambda^2\mathbb{E}Y^2/2$, the second-order expansion of the growth rate $\mathbb{E}\log(1+\lambda Y)$ (Kelly, 1956; Waudby-Smith and Ramdas, 2024). It affects power only, never validity.

Proposition 4.4 (Exact-interval confidence sequence)**.** *Let* X_i *be i.i.d. with mean* μ*, and let* $\lambda_i \in [0, b]$*,* $b < 1/2$*, be predictable and not depend on the candidate mean* m*. Set* $K_t^+(m) = \prod_{i\leq t}(1 + \lambda_i(X_i - m))$*,* $K_t^-(m) = \prod_{i\leq t}(1 + \lambda_i(m - X_i))$ *and*

$$C_t = \{m \in [-1, 1] : K_t^+(m) < 2/\alpha \text{ and } K_t^-(m) < 2/\alpha\}.$$

Then each C_t *is an interval,* $\mathbb{P}(\forall t : \mu \in C_t) \geq 1 - \alpha$*, and the same holds for the running intersection* $\bigcap_{s\leq t} C_s$*.* *[adaptation of Waudby-Smith and Ramdas, 2024]*

Algorithm 1 Paired regression or certification test (default bet: aGRAPA)

Require: cap c, null mean m, side s, level α, max bet $b < 1$, prior second moment $v_0 = 1/4$

$\log K \leftarrow 0$, $S_1 \leftarrow 0$, $S_2 \leftarrow 0$, $n \leftarrow 0$
for each case i as it resolves **do**
  $\lambda \leftarrow \min\{\max\{(S_1/(n+1))/((v_0+S_2)/(n+1)), 0\},\ b/(1+sm)\}$ ▷ uses past data only
  $Y \leftarrow s((C_i - B_i)/c - m)$; $\log K \leftarrow \log K + \log(1+\lambda Y)$
  $S_1 \leftarrow S_1 + Y$, $S_2 \leftarrow S_2 + Y^2$, $n \leftarrow n+1$
  if $\log K \geq \log(1/\alpha)$ **then return** reject H_0 at case n
  end if
end for
return do not reject

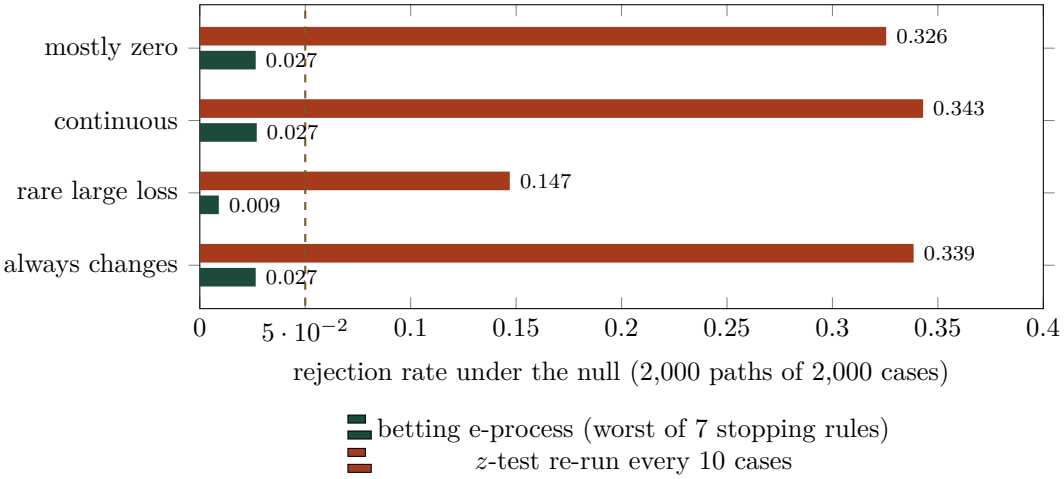


Figure 1: Continuous monitoring at the null boundary, $\alpha = 0.05$ (dashed). Synthetic; `scripts/math_sequential.py`.

Proof. For $m \in [-1, 1]$ every factor $1 + \lambda_i(X_i - m) \geq 1 - 2b > 0$ and is non-increasing in m; a product of positive non-increasing functions is non-increasing, so $\{m : K_t^+(m) \geq 2/\alpha\}$ is an initial segment of $[-1, 1]$. Symmetrically $\{m : K_t^-(m) \geq 2/\alpha\}$ is a final segment, so C_t is an interval. At $m = \mu$, $\mathbb{E}[1 + \lambda_i(X_i - \mu) \mid \mathcal{F}_{i-1}] = 1$, so $K_t^\pm(\mu)$ are non-negative martingales with initial value 1; by Lemma 4.1 each exceeds $2/\alpha$ at some time with probability at most $\alpha/2$, and by a union bound $\mu \in C_t$ for all t with probability at least $1 - \alpha$. On that event μ lies in every C_s and so in their intersection. □

Evidence (synthetic). Nulls at the boundary $\mu = m = 0$, four null shapes, seven stopping rules applied to the same 2,000 paths of 2,000 cases (including "stop when it looks bad"): the worst rejection rate of the betting test was 0.027 at $\alpha = 0.05$. A one-sided z-test re-run every 10 cases rejected 0.147 to 0.343 of the time (Figure 1). Power (ternary data, 10% of cases changed) is in Figure 2.

5 Stratified guarantees for rare case types

Cases arrive in one stream; each carries a type $k \in \{1, \ldots, K\}$ (dispute family) observed before its outcome. For each type, $H_{0,k} : \mathbb{E}[X \mid \text{past}, \text{type} = k] \leq m_k$, tested by an e-process E_k that updates only on type-k cases.

Lemma 5.1 (Stratum e-processes on a shared stream)**.** *Under $H_{0,k}$, E_k is a non-negative*

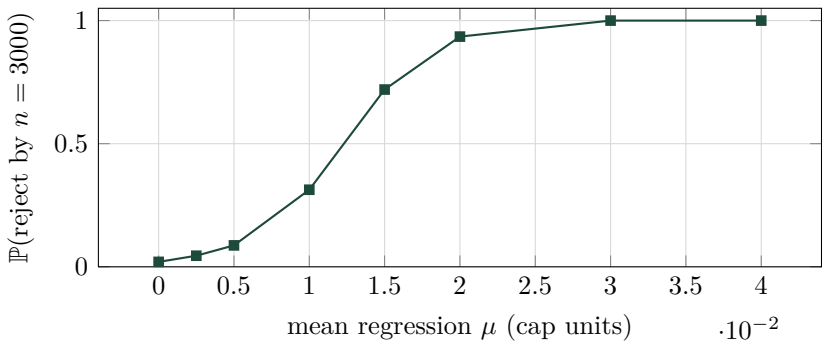


Figure 2: Power of the regression test (aGRAPA), ternary data with 10% of cases changed by ± 0.8, 1,000 runs per point. Synthetic.

supermartingale for the global filtration, so $\mathbb{P}(\sup_t E_{k,t} \geq a) \leq 1/a$ *and* $\mathbb{E}[E_{k,\tau}] \leq 1$ *for every global stopping time* τ. *[adaptation]*

Proof. Let $\mathcal{F}'_{t-1}$ be the global past together with the type of case t. If case t has type k, the factor is $1 + \lambda Y$ with λ $\mathcal{F}_{t-1}$-measurable and $\mathbb{E}[1 + \lambda Y \mid \mathcal{F}'_{t-1}] \leq 1$ by the argument of Proposition 4.3 applied conditionally on the type; otherwise the factor is 1. Taking conditional expectations given $\mathcal{F}_{t-1}$ preserves the inequality, and Lemma 4.1 finishes. □

Proposition 5.2 (Time-uniform weighted e-Bonferroni)**.** *Let* $w_k \geq 0$ *with* $\sum_k w_k = K$. *Flag type* k *the first time* $E_{k,t} \geq K/(\alpha w_k)$ *(never if* $w_k = 0$*). Then* $\mathbb{P}$(*some true null is ever flagged*) $\leq \alpha$, *under any dependence between types. [standard]*

Proof. By Lemma 5.1 with $a = K/(\alpha w_k)$ and a union bound over the set I_0 of true nulls, the probability is at most $\sum_{k \in I_0} \alpha w_k / K \leq \alpha$. □

Proposition 5.3 (Closed testing with mean e-values)**.** *At a stopping time* τ, *reject* $H_{0,k}$ *iff* $\frac{1}{|S|} \sum_{j \in S} E_{j,\tau} \geq 1/\alpha$ *for every* $S \ni k$. *Then the family-wise error is at most* α*; every hypothesis rejected by unweighted e-Bonferroni at level* α *is rejected; and for each size* s *the minimum over* $|S| = s$, $S \ni k$, *is attained by* k *together with the* $s - 1$ *smallest other e-values.*
[standard; Marcus et al., 1976; Vovk and Wang, 2021]

Proof. Let I_0 be the true nulls. The mean $\bar{E}_{I_0} = \frac{1}{|I_0|} \sum_{j \in I_0} E_{j,\tau}$ has $\mathbb{E}\bar{E}_{I_0} \leq 1$ by Lemma 5.1, so $\mathbb{P}(\bar{E}_{I_0} \geq 1/\alpha) \leq \alpha$ by Markov's inequality. A false rejection of some $k \in I_0$ requires the condition for $S = I_0 \ni k$, i.e. the event above. If $E_k \geq K/\alpha$ then for every $S \ni k$, $\frac{1}{|S|} \sum_S E_j \geq E_k/|S| \geq E_k/K \geq 1/\alpha$. For fixed size, the sum over S with $k \in S$ is E_k plus a sum of $s - 1$ others, minimised by the smallest ones. □

Proposition 5.4 (Weighted e-BH)**.** *At a stopping time, with* $e'_k = w_k E_{k,\tau}$, *reject the* r *largest* e'_k, *where* r *is the largest* j *with* $e'_{(j)} \geq K/(\alpha j)$. *Then* FDR $\leq \alpha \sum_{k \in I_0} w_k / K \leq \alpha$ *under any dependence. [standard; Wang and Ramdas, 2022]*

Proof. With R rejections, every rejected k satisfies $e'_k \geq K/(\alpha R)$, so $\mathbf{1}\{k \text{ rejected}\}/\max(R, 1) \leq \alpha e'_k / K$. Summing over $k \in I_0$ and taking expectations, FDR $\leq \frac{\alpha}{K} \sum_{k \in I_0} w_k \mathbb{E}[E_{k,\tau}] \leq \frac{\alpha}{K} \sum_{k \in I_0} w_k$. □

Remark 5.5. Propositions 5.3 and 5.4 hold at one stopping time; "reject whenever e-BH has ever rejected" during monitoring is not covered. The gate uses Proposition 5.2 while monitoring.

Evidence (synthetic). Eight types with frequencies $(0.30, 0.20, 0.15, 0.12, 0.10, 0.08, 0.04, 0.01)$; the rare type regresses by $+0.35$ cap units while every other type improves by 0.01, so the pooled mean improves (-0.0064). Over 500 runs of 8,000 cases a pooled anytime test flagged a regression 0.006 of the time; stratified e-Bonferroni flagged the rare type 1.000 of the time. At the null boundary the old zero-tolerance "no slice mean may get worse" check false-blocked 0.73 to 0.996 of the time, the stratified procedures at most 0.03.

6 Planning and the cap tax

Normalise $m = t_k/c_k$, $d = \Delta_k/c_k$, $\sigma = \sigma_k/c_k$; let $Y = X - m$ with $\mathbb{E}Y = d$, $M_2 = \mathbb{E}Y^2 = \sigma^2 + d^2$, constant bet λ and $a = \lambda(1+m) < 1$.

Lemma 6.1 (Growth bound)**.** *For $a \in [0,1)$ and $x \geq -a$, $\log(1+x) \geq x - \frac{x^2}{2(1-a)}$. Hence $g(\lambda) := \mathbb{E}\log(1+\lambda Y) \geq \lambda d - \frac{\lambda^2 M_2}{2(1-a)}$.* *[adaptation]*

Proof. Let $f(x) = \log(1+x) - x + \frac{x^2}{2(1-a)}$. Then $f(0) = 0$ and $f'(x) = x(\frac{1}{1-a} - \frac{1}{1+x})$. For $x \in [-a, 0]$, $1 + x \geq 1 - a$ so the bracket is ≥ 0 and $f' \leq 0$; for $x \geq 0$ both factors are ≥ 0. So f is minimised at 0 and $f \geq 0$. Apply with $x = \lambda Y \geq -\lambda(1+m) = -a$ and take expectations. □

Proposition 6.2 (Sufficient stratum sample size)**.** *Let $Z = \log(1+\lambda Y)$, $v = \lambda^2 M_2/(1-a)^2$, $b = \lambda d - \log(1-a)$, $l = \log(1/\beta)$ and $\varepsilon(n) = lb/3 + \sqrt{(lb/3)^2 + 2lnv}$. For i.i.d. observations, the constant-bet e-test at level α_k rejects by n observations with probability at least $1-\beta$ whenever*

$$n\, g(\lambda) - \varepsilon(n) \geq \log(1/\alpha_k). \tag{4}$$

If additionally the type count is $n_k \sim \mathrm{Binomial}(N, \pi_k)$, independent of the type-k outcomes, and $\mathbb{P}(n_k \geq n) \geq 1-\gamma$, the power with N total cases is at least $(1-\beta)(1-\gamma)$. *[adaptation]*

Proof. For $x \geq -a$, $|\log(1+x)| \leq |x|/(1-a)$: for $x \geq 0$, $\log(1+x) \leq x$; for $x \in [-a, 0)$, $-\log(1+x) \leq -x/(1+x) \leq |x|/(1-a)$. Hence $\mathrm{Var}Z \leq \mathbb{E}Z^2 \leq v$. By Jensen, $\mathbb{E}Z \leq \log(1+\lambda d) \leq \lambda d$, and $Z \geq \log(1-a)$, so $\mathbb{E}Z - Z \leq b$. Bernstein's inequality (Boucheron et al., 2013, Thm. 2.10 and Cor. 2.11) for $S_n = \sum_{i \leq n} Z_i$ gives $\mathbb{P}(n\mathbb{E}Z - S_n \geq \varepsilon) \leq \exp\big(-\frac{\varepsilon^2}{2(nv + b\varepsilon/3)}\big)$, and the stated $\varepsilon(n)$ solves $\varepsilon^2 = 2l(nv + b\varepsilon/3)$, so the right side is β. With probability at least $1-\beta$, $\log K_n = S_n \geq n\mathbb{E}Z - \varepsilon(n) \geq ng(\lambda) - \varepsilon(n) \geq \log(1/\alpha_k)$ by Lemma 6.1 and (4), and the test rejects. For the second claim, the test rejects if its wealth ever crosses, so its rejection probability given $n_k \geq n$ is at least its probability of crossing by the n-th type-k case, which is at least $1-\beta$ by independence of counts and outcomes. □

Proposition 6.3 (The cap tax: a lower bound for every valid test)**.** *Let $P_1 = \delta_{m+d}$ and $P_0 = (1-p)\delta_{m+d} + p\,\delta_{-1}$ with $p = d/(1+m+d)$. Every test valid at level α for all laws on $[-1, 1]$ with mean $\leq m$ and with power $1-\beta$ against P_1 after n i.i.d. observations satisfies*

$$n \;\geq\; \frac{\mathrm{kl}(1-\beta \| \alpha)}{\log\big(1 + \frac{d}{1+m}\big)}, \tag{5}$$

and a sequential test satisfies the same bound for $\mathbb{E}_{P_1}[\tau]$. *[standard; Kaufmann et al., 2016]*

Proof. P_0 has mean $(1-p)(m+d) - p = m$, so it is in the null and $P_0^n(A) \leq \alpha$ for the rejection event A, while $P_1^n(A) \geq 1-\beta$. $\mathrm{KL}(P_1 \| P_0) = \log\frac{1}{1-p} = \log\frac{1+m+d}{1+m}$, and $\mathrm{KL}(P_1^n \| P_0^n) = n\mathrm{KL}(P_1 \| P_0)$. Lemma 4.2 gives the bound. The sequential version replaces n by $\mathbb{E}_{P_1}[\tau]$ via the change-of-measure lemma of Kaufmann et al. (2016, Lemma 1). □

Algorithm 2 Grid mixture of constant bets

Require: m, s, α, $\lambda_{\max} = 0.95/(1+sm)$, ratio $\rho = 1/2$, grid size $J = 14$

$\lambda_j \leftarrow \lambda_{\max}\rho^j$ and $L_j \leftarrow 0$ for $j = 0, \dots, J-1$

for each case i **do**

  $Y \leftarrow s(X_i - m)$; $L_j \leftarrow L_j + \log(1+\lambda_j Y)$ for all j

  $\log E \leftarrow \log\left(\frac{1}{J}\sum_j e^{L_j}\right)$ ▷ computed stably

  if $\log E \geq \log(1/\alpha)$ **then return** reject

  end if

end for

Because $d = \Delta/c$, the required sample size grows linearly with the declared cap: a loose cap is paid for in cases by every valid nonparametric test. On the synthetic illustration (cap \$100, regression \$5, s.d. \$20, 1% stratum, eight strata) the Bernstein plan needs 795 rare-stratum cases (simulated minimum about 341) and the optimised α allocation saves 28% to 35% of total traffic against uniform α/K.

7 Variance-adaptive betting and a floor from observed data

Paired cost differences are mostly zero with a few large values. The default bet of Algorithm 1 caps λ at half the largest valid bet, which binds when the variance is small relative to the mean, and its plug-in mean is noisy while most observations are zero. We compare four bets, all valid for the same null: *agrapa* (default), *agrapa_wide* (cap $0.95/(1+sm)$), *plugin* (the predictable plug-in bet of Waudby-Smith and Ramdas, 2024, $\lambda_t = \min\{0.95/(1+sm), \sqrt{2\log(1/a)/(\hat{\sigma}^2_{t-1} t\log(1+t))}\}$) and *mixture* (Algorithm 2).

Proposition 7.1 (Grid mixture)**.** *With $\lambda_j = \lambda_{\max}\rho^j$, $j < J$, $\lambda_{\max} \leq b/(1+sm)$, $b < 1$, the process $E_t = \frac{1}{J}\sum_j \prod_{i\leq t}(1+\lambda_j Y_i)$ is a non-negative supermartingale with $E_0 = 1$ under H_0, and on every path*

$$\log E_t \;\geq\; \max_j \log K_t(\lambda_j) - \log J. \tag{6}$$

If the Y_i are i.i.d. and $g(\lambda) = \mathbb{E}\log(1+\lambda Y)$ is maximised at $\lambda^\star \in [\lambda_{J-1}, \lambda_0]$, then $\liminf_t \frac{1}{t}\log E_t \geq \rho\, g(\lambda^\star)$ almost surely. *[adaptation]*

Proof. Each $K_t(\lambda_j)$ is a non-negative supermartingale with $K_0 = 1$ (Proposition 4.3 with a constant bet). A convex combination of non-negative supermartingales on the same filtration is a non-negative supermartingale, with initial value 1. A sum of non-negative terms is at least each term, so $E_t \geq K_t(\lambda_j)/J$ for every j, which is (6). For the rate: by the strong law, $\frac{1}{t}\log K_t(\lambda_j) \to g(\lambda_j)$ a.s. for each of the finitely many j (the summands are bounded below by $\log(1-b)$ and above by $\log 2$); with (6), $\liminf \frac{1}{t}\log E_t \geq \max_j g(\lambda_j)$. Some λ_j lies in $[\rho\lambda^\star, \lambda^\star]$. g is concave with $g(0) = 0$, so $g(\theta\lambda^\star) \geq \theta g(\lambda^\star)$ for $\theta \in [0,1]$, and $g(\lambda_j) \geq \rho g(\lambda^\star)$. □

Proposition 7.2 (The best constant-bet growth is a distance to the null)**.** *Let P be a law on $[-1,1]$ with mean above m, $Y = X - m$, $\bar{\lambda} = 1/(1+m)$, and $g(\lambda) = \mathbb{E}_P\log(1+\lambda Y)$ on $[0, \bar{\lambda}]$. Let $\lambda^\star$ maximise g with $g^\star = g(\lambda^\star) > 0$, and suppose either (i) $\lambda^\star < \bar{\lambda}$, or (ii) $\lambda^\star = \bar{\lambda}$ and $P([-1, -1+\varepsilon_0)) = 0$ for some $\varepsilon_0 > 0$. Define*

$$Q = \frac{1}{1+\lambda^\star Y}\,P + w\,\delta_{-1}, \qquad w = 1 - \mathbb{E}_P\Big[\frac{1}{1+\lambda^\star Y}\Big].$$

Then $w \geq 0$ ($w = 0$ in case (i)), Q is a probability on $[-1,1]$ with mean exactly m, and $\mathrm{KL}(P\|Q) = g^\star$. Consequently every test valid at level α for all laws on $[-1,1]$ with mean $\leq m$ and with power $1-\beta$ against P after n i.i.d. observations satisfies

$$n \;\geq\; \mathrm{kl}(1-\beta\|\alpha)/g^\star, \tag{7}$$

and a sequential test satisfies the same bound for $\mathbb{E}_P[\tau]$.

[adaptation; $g^\star = K_{\inf}$ *is the duality of Honda and Takemura, 2010]*

Proof. Write $c(\lambda) = \mathbb{E}_P[Y/(1+\lambda Y)]$. In case (i), on a neighbourhood $[0, \lambda^\star + \delta]$ with $\lambda^\star + \delta < \bar{\lambda}$, $1 + \lambda Y \geq 1 - (\lambda^\star + \delta)(1+m) > 0$, so $Y/(1+\lambda Y)$ is bounded and g is differentiable there with $g' = c$ (dominated convergence). g is concave, so an interior maximiser, or the maximiser $\lambda^\star = 0$ (excluded since $g^\star > 0 = g(0)$), has $c(\lambda^\star) = 0$. In case (ii), $1 + \bar{\lambda} Y = (1+X)/(1+m) \geq \varepsilon_0/(1+m)$ P-a.s., so g is differentiable on $[0, \bar{\lambda}]$ (one-sided at $\bar{\lambda}$) with $g' = c$, and a maximum at the right end forces $c(\bar{\lambda}) \geq 0$. In both cases $c(\lambda^\star) \geq 0$ and $\lambda^\star c(\lambda^\star)(1 - \lambda^\star(1+m)) = 0$.

Since $\frac{1}{1+\lambda Y} = 1 - \frac{\lambda Y}{1+\lambda Y}$, $\mathbb{E}_P\big[\frac{1}{1+\lambda^\star Y}\big] = 1 - \lambda^\star c(\lambda^\star)$, so $w = \lambda^\star c(\lambda^\star) \geq 0$, with $w = 0$ in case (i), and Q has total mass 1; its density with respect to P is positive, so Q is a probability. Its mean shift is

$$\mathbb{E}_Q[Y] = \mathbb{E}_P\Big[\frac{Y}{1+\lambda^\star Y}\Big] + w \cdot (-(1+m)) = c(\lambda^\star)(1 - \lambda^\star(1+m)) = 0,$$

using $c(\lambda^\star) = 0$ in case (i) and $\lambda^\star(1+m) = 1$ in case (ii). So $\mathbb{E}_Q X = m$ and Q lies in the null. P puts no mass at -1 in case (ii) and $w = 0$ in case (i), so on the support of P, $\frac{dP}{dQ} = 1 + \lambda^\star Y$ and $\mathrm{KL}(P\|Q) = \mathbb{E}_P \log(1 + \lambda^\star Y) = g^\star$. For the rejection event A, $P^n(A) \geq 1 - \beta$ and $Q^n(A) \leq \alpha$; Lemma 4.2 and $\mathrm{KL}(P^n\|Q^n) = n\, g^\star$ give (7). The sequential statement follows from Kaufmann et al. (2016, Lemma 1). □

Remark 7.3. The converse inequality $\mathrm{KL}(P\|Q') \geq g^\star$ for every null Q' (so that $g^\star$ is exactly the smallest divergence to the null, $K_{\inf}$) follows from the Donsker–Varadhan variational formula with the test function $\log(1 + \lambda^\star Y)$; we only use the direction proved above. Applied to an empirical law $\hat{P}$, (7) answers: if cases were drawn i.i.d. from the paired costs we observed, how many would any valid test need? It is a statement about the observed distribution, not about a population.

Evidence on E1's planted regressions (synthetic). We replayed the E1 policies on all 10,000 benchmark cases and stored the paired merchant-cost differences (USD minor units, no case content; file `results/experiments/E1_paired_costs.json`). The gate is the E1 detector's block family (six families plus pooled, weighted e-Bonferroni on running maxima, cap \$200, $\alpha = 0.05$), with each betting strategy in turn. With the default bet it reproduces E1's verdicts exactly (R2 blocked on the pooled process; R3 on buyer's remorse, service failure and pooled; the rest held). Table 2 and Figure 3 summarise `scripts/math_betting.py analyse`.

Table 2: Betting strategies on E1's paired costs (synthetic). Top: largest running-max $\log E$ over the seven gate hypotheses on the full benchmark (rejection needs $\log(7/0.05) = 4.94$). Bottom: $\mathbb{P}$(gate blocks) when n cases are resampled i.i.d. from the benchmark's paired costs, 100 resamples per cell. All four bets are valid for the same null; at the null boundary (centred R5 and R3 costs, 400 runs of 2,000) they rejected 0 to 0.0025 of the time.

		agrapa (default)	agrapa_wide	plugin	mixture
$\log E$, full benchmark	R1	2.13	3.54	2.84	1.54
	R2	9.91	16.62	7.93	14.23
	R3	34.42	58.78	15.84	56.71
	R4	1.96	3.56	3.66	1.54
	R5	2.91	**5.11**	4.62	2.80
$\mathbb{P}$(block), resampled	R2, $n = 2{,}500$	0.00	0.19	0.00	0.00
	R2, $n = 5{,}000$	0.37	0.91	0.43	0.64
	R1, $n = 10{,}000$	0.00	0.17	0.00	0.01
	R4, $n = 10{,}000$	0.00	0.04	0.01	0.00
	R5, $n = 10{,}000$	0.00	0.43	0.39	0.02

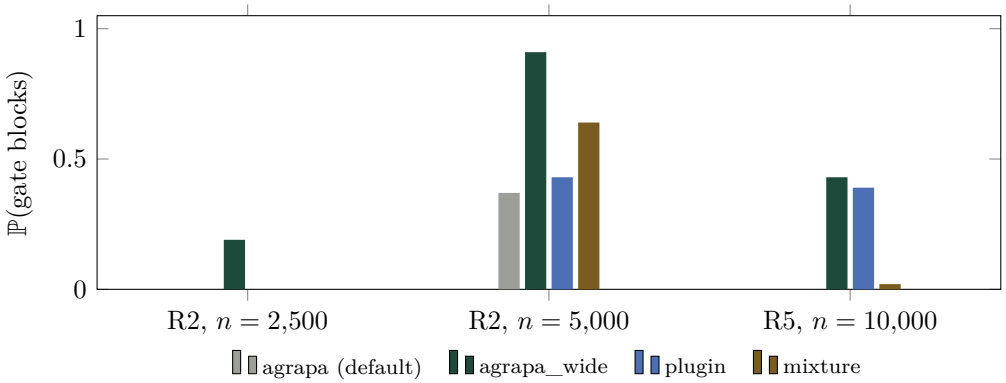


Figure 3: Power at equal validity on E1's planted regressions, cases resampled i.i.d. from the benchmark's paired costs (synthetic).

What the numbers say:

- **The default bet's cap is the bottleneck.** In almost every regressed family the best constant bet on the observed costs is the largest valid one, $\lambda^\star = 1/(1+m)$ (case (ii) of Proposition 7.2): the differences are zero on most cases and large and positive on a few, so the Kelly bet is everything. The default stops at half of it. Raising the cap to 0.95 multiplies the final log-wealth by 1.66 to 1.82 and lifts R2's resampled power at 5,000 cases from 0.37 to 0.91.
- **R5, the regression E1 missed, is visible to a stratified test, not to a pooled one.** Its pooled mean is 1.34 USD minor per case, about one cent ($t = 0.27$), and Proposition 7.2 says any valid pooled test needs at least 509,153 cases for power 0.8 at $\alpha = 0.05$. But its cost is concentrated: in buyer's remorse disputes the mean (after clipping at the cap) is 64.35 minor per case, and the floor there is 689 family cases, about 3,486 benchmark cases in total (6,316 at the gate's per-hypothesis level 0.05/7). On the actual benchmark the wide bet blocked R5 on buyer's remorse ($\log E = 5.11$); on resampled benchmarks of 10,000 cases it did so 43% of the time, the plug-in bet 39%, the default never.
- **The mixture pays for its grid.** Its log-wealth is within $\log 14 = 2.64$ of the best grid bet (Proposition 7.1), which costs it detections when the best bet is the largest one. It wins only when $\lambda^\star$ is far from the cap, which these regressions are not.
- **Nothing helps on 500-case folds.** No strategy blocked any of the 20 folds of any policy; the floors above are thousands of cases.

These are statements about the benchmark's own paired costs (resampling reuses them), not about a population of agent changes. A separate workstream is adding control variates to `scruple/stats`; that work was not merged when this was written, and nothing here depends on it.

8 Asymmetric costs

Let θ be the true mean per-case loss difference and δ the tolerance. A wrong approval (promote while $\theta > \delta$) costs c_A; a wrong block costs c_B.

Proposition 8.1 (Bayes rule)**.** *Under a posterior* Π*, the rule that minimises posterior expected 0–1 loss promotes iff* $\Pi(\theta > \delta) \le c_B/(c_A + c_B)$*.* *[standard; Berger, 1985]*

Proof. Posterior risk of promoting is $c_A\Pi(\theta > \delta)$ and of blocking $c_B\Pi(\theta \le \delta) = c_B(1 - \Pi(\theta > \delta))$; promote iff the first is at most the second, which rearranges to the threshold. Bayes risk is the

integral of posterior risk over the marginal of the data (Fubini), so the pointwise minimiser minimises it. □

Proposition 8.2 (Composition with the certificate)**.** *Let E be an e-value (or a running maximum used with Ville) for $H_0 : \theta \geq \delta$ and $A = \{E \geq 1/\alpha\}$. The composed rule promotes iff $x \in A$ and the Bayes rule promotes. Then (1) $\mathbb{P}_\theta(\text{promote}) \leq \alpha$ for every $\theta \geq \delta$, whatever the prior and whether or not the posterior model is right; (2) among all rules $\varphi \leq \mathbf{1}_A$ it minimises Bayes risk under the model. [adaptation; restricted Bayes in the sense of Hodges and Lehmann, 1952]*

Proof. (1) {promote} $\subseteq A$ and $\mathbb{P}_\theta(A) \leq \alpha$ by Markov's inequality or Lemma 4.1. (2) Bayes risk equals $\int [\varphi(x) r_P(x) + (1 - \varphi(x)) r_B(x)]\, dm(x)$, with r_P, r_B the posterior risks of promoting and blocking. Under the constraint $\varphi \leq \mathbf{1}_A$, the integrand is minimised pointwise by $\varphi(x) = \mathbf{1}\{x \in A, r_P(x) \leq r_B(x)\}$, which is the composed rule. □

Synthetic evidence: at 1:1 costs the Bayes rule alone promotes 0.614 of boundary-harmful changes; composed with the certificate, 0.0017.

9 What the whole promote decision guarantees

The gate promotes iff every conjunct passes: hard checks H on every case, a soft-score threshold S, no block from the block family B (Proposition 5.2), and the certificate T (Proposition 4.3 with $s = -1$). A *harm* is a property of the candidate we never want to ship. A conjunct c *covers* harm j at level $\alpha_{c,j}$ if $\sup\{\mathbb{P}(c \text{ passes}) : \text{candidate has harm } j\} \leq \alpha_{c,j}$; a deterministic check that reads the harm on every case has level 0, and a sampled audit has the level of Theorem 10.3.

Theorem 9.1 (Composition)**.** *(a) (Promote side.) For a candidate with harms J,*

$$\mathbb{P}(\textit{promote}) \leq \min_{j \in J} \min_{c} \alpha_{c,j},$$

and the bound is 1 if no conjunct covers any harm in J.

(b) (Block side.) If the candidate has no harm and its hard checks raise a false alarm on case i with probability ϕ_i, then $\mathbb{P}(\textit{block}) \leq \alpha_{\textit{block}} + \mathbb{P}(\exists i : \textit{false alarm on } i)$, which equals $\alpha_{\textit{block}} + 1 - \prod_i (1 - \phi_i)$ under independence and is at most $\alpha_{\textit{block}} + \sum_i \phi_i$ in general.

(c) (Resubmission.) If submission k of a candidate line is certified at level α_k with $\sum_k \alpha_k \leq \alpha$, the probability that any harmful submission is ever promoted is at most α.

[standard, applied; intersection-union principle of Berger, 1982; spending in the spirit of Foster and Stine, 2008]

Proof. (a) {promote} $\subseteq \{c \text{ passes}\}$ for every conjunct c, so $\mathbb{P}(\text{promote}) \leq \mathbb{P}(c \text{ passes}) \leq \alpha_{c,j}$ for each covering pair; take the minimum. (b) {block} $\subseteq \{B \text{ fires}\} \cup \{\text{some hard check fires}\}$; the first has probability at most α_{block} for a harmless candidate (Proposition 5.2); the second is $1 - \prod_i (1 - \phi_i)$ under independence and at most $\sum_i \phi_i$ by a union bound. (c) The event is the union over k of "submission k is harmful and promoted", each of probability at most α_k by (a). □

Corollary 9.2 (Per-type certificates)**.** *Run one certificate e-process per case type for $H_{0,k} : \mu_k \geq t_k$ on the interleaved stream and promote only if every type is certified at level α. If any one type is harmful, $\mathbb{P}(\textit{promote}) \leq \alpha$, with no division of α by K. [adaptation]*

Proof. If type k is harmful, "type k certified" is a conjunct covering that harm at level α by Lemma 5.1; apply Theorem 9.1(a). □

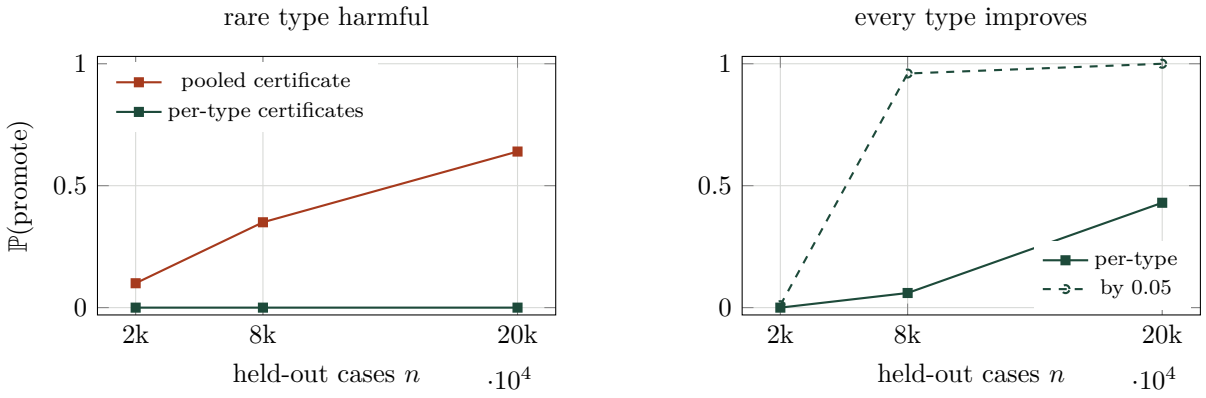


Figure 4: Left: a 1% type regresses by 0.35 cap units and the rest improve by 0.01. The pooled certificate promotes more often as n grows, because the pooled mean really is below tolerance; per-type certificates never did. Right: the power cost of per-type certificates (the pooled certificate promotes 1.00 in both settings). 100 runs per point; synthetic; `scripts/math_composition.py`.

What is not guaranteed. (1) Harms no conjunct covers: E2's cheaper fraud (Section 11.2) makes the null "cost is worse" false, so B and T say nothing; E4's unfair wins are scored by the soft OUT-01. (2) Rare-type regressions under the shipped *pooled* certificate (Figure 4). (3) Soft scores and the Bayes rule can only remove promotions, but carry no guarantee of their own. (4) Level 0 for a hard check assumes the check is correct; the verifier and the simulator share a world model. (5) Shift between held-out and live traffic, and delayed outcomes in canary.

Evidence (synthetic). With one common type exactly at its tolerance, per-type certificates promoted at most 0.02 of the time ($\alpha = 0.05$). A hard check with false-alarm rate $\phi = 0.0019$ per case (the honest policy's measured INV-01 rate) blocked an honest candidate 0.319 of the time at $n = 200$, 0.596 at 500, 0.979 at 2,000 and 1.000 at 10,000 (bounds 0.366, 0.664, 1.000, 1.000). Twenty harmful versions at a fixed $\alpha = 0.05$: one was promoted in 0.402 of runs; with spending $\alpha_k = 6\alpha/(\pi^2 k^2)$, in 0.017.

10 Hacking-gap bounds

Definition 10.1 (Hacking gap)**.** On case i the candidate has a proxy score P_i (a self-report or a naive win label) and an anchored score A_i (recomputed from the ledger), in cap units, with $0 \le D_i = P_i - A_i \le \varepsilon$. The hacking gap is $\Gamma = \frac{1}{n}\sum_{i \le n} D_i$. Fix a tolerance $\eta \in [0, \varepsilon]$; case i is *hacked* if $D_i > \eta$, $h_i = \mathbf{1}\{D_i > \eta\}$ and $M_t = \sum_{i \le t} h_i$. Some cases are *flagged*; the batch is *accepted* iff no case is flagged.

Assumption 10.2 (Catch floor)**.** Cases are handled in order. Before case i the agent holds information $\mathcal{G}_i \supseteq \mathcal{F}_{i-1}$ (the history, any signal about case i, its own randomness) and chooses D_i as a $\mathcal{G}_i$-measurable variable. Let $\mathcal{H}_i = \sigma(\mathcal{G}_i)$ and Fl_i the event that case i is flagged, with $\mathrm{Fl}_i \in \mathcal{F}_i$. There is $\rho \in [0, 1]$ with

$$\mathbb{P}(\mathrm{Fl}_i \mid \mathcal{H}_i) \;\ge\; \rho \quad \text{on } \{h_i = 1\}.$$

Assumption 10.2 allows randomised and adaptive policies, audits that depend on the past, an agent that sees past audit results or partial signals, and correlated detections; flags on honest cases are allowed.

Theorem 10.3 (Gap of an accepted change, adaptive agents)**.** *Under Assumption 10.2, with* $m_0 = \lceil n(\gamma-\eta)/\varepsilon \rceil$,

$$\mathbb{P}(\textit{accepted and } \Gamma \geq \gamma) \leq (1-\rho)^{m_0}, \tag{8}$$

$$\mathbb{E}[\Gamma\, \mathbf{1}\{\textit{accepted}\}] \leq \eta + \frac{\varepsilon}{n\, e\, \ln \frac{1}{1-\rho}}. \tag{9}$$

[adaptation of the survival argument to a supermartingale; framing candidate novel]

Proof. Write $r = 1-\rho$; if $\rho = 0$ both bounds are trivial ($\Gamma \leq \varepsilon$), and if $\rho = 1$ no hacked case is accepted, so assume $0 < \rho < 1$. Let $A_t = \bigcap_{i \leq t} \mathrm{Fl}_i^c$ and $Z_t = \mathbf{1}_{A_t} r^{-M_t}$, $Z_0 = 1$. Then $Z_i = Z_{i-1}\, r^{-h_i} \mathbf{1}_{\mathrm{Fl}_i^c}$, and Z_{i-1} and h_i are $\mathcal{H}_i$-measurable, so

$$\mathbb{E}[Z_i \mid \mathcal{H}_i] = Z_{i-1} r^{-h_i}\, \mathbb{P}(\mathrm{Fl}_i^c \mid \mathcal{H}_i) \leq Z_{i-1} :$$

if $h_i = 1$, $\mathbb{P}(\mathrm{Fl}_i^c \mid \mathcal{H}_i) \leq 1-\rho = r$ by Assumption 10.2; if $h_i = 0$ the factor is at most 1. By the tower property ($\mathcal{F}_{i-1} \subseteq \mathcal{H}_i$), $\mathbb{E}[Z_i] \leq \mathbb{E}[Z_{i-1}]$ and $\mathbb{E}[Z_n] \leq 1$. On A_n, $Z_n = r^{-M_n}$, so

$$\mathbb{P}(A_n \cap \{M_n \geq m_0\}) = \mathbb{E}\big[Z_n\, r^{M_n} \mathbf{1}\{M_n \geq m_0\}\big] \leq r^{m_0} \mathbb{E}[Z_n] \leq r^{m_0}.$$

Since $D_i \leq \eta$ for unhacked and $D_i \leq \varepsilon$ for hacked cases, $\Gamma \leq \frac{(n-M_n)\eta + M_n \varepsilon}{n} \leq \eta + \frac{\varepsilon M_n}{n}$, so $\Gamma \geq \gamma$ forces $M_n \geq n(\gamma-\eta)/\varepsilon$, i.e. $M_n \geq m_0$; this proves (8). For (9), $\mathbb{E}[M_n \mathbf{1}_{A_n}] = \mathbb{E}[Z_n M_n r^{M_n}] \leq \sup_{x \geq 0} x r^x \cdot \mathbb{E}[Z_n]$. The function $x \mapsto x r^x$ has derivative $r^x(1 + x \ln r)$, vanishing at $x = 1/\ln(1/r)$ where its value is $1/(e \ln(1/r))$. Hence $\mathbb{E}[\Gamma \mathbf{1}_{A_n}] \leq \eta + \frac{\varepsilon}{n} \mathbb{E}[M_n \mathbf{1}_{A_n}] \leq \eta + \frac{\varepsilon}{ne \ln(1/r)}$. □

Corollary 10.4 (Independent audits)**.** *If each case is audited independently with probability* q*, independently of the agent's information, and an audited hacked case is flagged with probability at least* $1-\beta$ *given the agent's information, then Theorem 10.3 holds with* $\rho = q(1-\beta)$*. For* $\beta = 0$ *this is the earlier spot-check bound:* $\mathbb{E}[\Gamma \mathbf{1}\{acc\}] \leq \eta + \varepsilon/(ne \ln \frac{1}{1-q}) \approx \eta + \varepsilon/(enq)$.

Proof. $\mathbb{P}(\mathrm{Fl}_i \mid \mathcal{H}_i) \geq \mathbb{P}(\mathrm{audited}_i \mid \mathcal{H}_i)(1-\beta) = q(1-\beta)$ on $\{h_i = 1\}$. □

Corollary 10.5 (Leaky audits)**.** *Let case* i *be audited* ($a_i = 1$) *with probability* q*, independently of* $\mathcal{F}_{i-1}$*, let the agent observe a signal* S_i *with* $\mathbb{P}(S_i = s \mid a_i = 1, \mathcal{F}_{i-1}) \geq \ell\, \mathbb{P}(S_i = s \mid a_i = 0, \mathcal{F}_{i-1})$ *for every* s *and some* $\ell \in [0,1]$*, let the agent's choice depend only on* $(\mathcal{F}_{i-1}, S_i)$ *and independent randomness, and let an audited hacked case be flagged with probability at least* $1-\beta$ *given* $\mathcal{H}_i$ *and* $a_i = 1$*. Then Assumption 10.2 holds with*

$$\rho = (1-\beta)\, \frac{q\ell}{q\ell + 1 - q}. \tag{10}$$

The bound is attained: an agent that sees "audited" with probability $1-\ell$ *on audited cases, sees "not audited" otherwise, and hacks only on "not audited", has conditional catch probability exactly* ρ *on every hacked case when* $\beta = 0$*.* *[adaptation]*

Proof. By Bayes, $\mathbb{P}(a_i = 1 \mid S_i = s, \mathcal{F}_{i-1}) = \frac{qL(s)}{qL(s)+1-q}$ with $L(s) = \frac{\mathbb{P}(s \mid a_i=1, \mathcal{F}_{i-1})}{\mathbb{P}(s \mid a_i=0, \mathcal{F}_{i-1})} \geq \ell$, and $x \mapsto \frac{qx}{qx+1-q}$ is non-decreasing. Independent randomness does not change this posterior, so $\mathbb{P}(a_i = 1 \mid \mathcal{H}_i) \geq \frac{q\ell}{q\ell+1-q}$ and $\mathbb{P}(\mathrm{Fl}_i \mid \mathcal{H}_i) \geq (1-\beta)\mathbb{P}(a_i = 1 \mid \mathcal{H}_i)$ on $\{h_i = 1\}$. For the attaining agent, the signal "not audited" has $L = \ell/1 = \ell$, so the posterior equals $q\ell/(q\ell + 1 - q)$ exactly. □

Theorem 10.6 (Minimal audit rate with an imperfect verifier)**.** *Cases are audited independently at rate* q*; an audited hacked case is flagged with probability* $1-\beta$ *and an audited honest case with probability* ϕ*, independently, with* $q\phi < 1$*. An accepted candidate pays the optimiser* $V + \sum_{i \in S} g_i$ *(honest value* $V \geq 0$ *plus a gain* $0 \leq g_i \leq g$ *per hacked case in the set* S*), a rejected one pays*

$-C$ *($C \geq 0$), and* $K = V + C > 0$. *Let* $\mathrm{EU}(S)$ *be the expected payoff of hacking* S. *Then* $\mathrm{EU}(S) \leq \mathrm{EU}(\emptyset)$ *for every* S *if*

$$D := (1-\beta)(K+g) - \phi K > 0 \quad \textit{and} \quad q \;\geq\; q^\star := \frac{g}{D}, \tag{11}$$

and conversely, if some case offers gain exactly $g > 0$ *and* $\mathrm{EU}(S) \leq \mathrm{EU}(\emptyset)$ *for every* S, *then* (11) *holds.*

[candidate novel (framing); the deterrence condition goes back to Becker, 1968; Avenhaus et al., 2002]

Proof. With $|S| = m$, $\mathbb{P}(\mathrm{acc} \mid S) = (1-q\phi)^{n-m}(1-q(1-\beta))^m = (1-q\phi)^n s^m$ with $s = \frac{1-q(1-\beta)}{1-q\phi}$. Hence $\mathrm{EU}(S) + C = (1-q\phi)^n s^m (K + \sum_S g_i) \leq (1-q\phi)^n f(m)$ with $f(m) = s^m(K+mg)$, and $\mathrm{EU}(\emptyset) + C = (1-q\phi)^n f(0)$. Now

$$\frac{f(m+1)}{f(m)} = s\Big(1 + \frac{g}{K+mg}\Big) \leq s\Big(1+\frac{g}{K}\Big),$$

so f is non-increasing whenever $s(K+g) \leq K$. This inequality is $(1-q(1-\beta))(K+g) \leq K(1-q\phi)$, i.e. $g \leq q[(1-\beta)(K+g) - \phi K] = qD$, which under $D > 0$ is $q \geq g/D$. This proves sufficiency. Conversely, take $S = \{i\}$ with $g_i = g$: $\mathrm{EU}(\{i\}) \leq \mathrm{EU}(\emptyset)$ is $(1-q\phi)^{n-1}(1-q(1-\beta))(K+g) \leq (1-q\phi)^n K$; dividing by $(1-q\phi)^{n-1} > 0$ gives $g \leq qD$, which with $g > 0$ forces $D > 0$ and $q \geq g/D$. □

Corollary 10.7 (When no audit rate deters)**.** *Under the conditions of Theorem 10.6, some* $q \in [0,1]$ *deters every hack iff* $\beta g \leq (1-\beta-\phi)K$, *and* $q^\star$ *is increasing in* β *and in* ϕ. *If instead* $K = 0$ *(no honest value and nothing lost when caught, as with free resubmission), hacking one case is unprofitable only if* $q(1-\beta) = 1$*: spot checks cannot deter an agent that may resubmit for free.*

Proof. A deterring $q \leq 1$ exists iff $g \leq D$ (the condition $g \leq qD$ is easiest at $q = 1$), and $g \leq (1-\beta)(K+g) - \phi K$ rearranges to $\beta g \leq (1-\beta-\phi)K$. D decreases in β and in ϕ, so $q^\star = g/D$ increases. If $K = 0$, $\mathrm{EU}(\{i\}) + C = (1-q\phi)^{n-1}(1-q(1-\beta))g$, which exceeds $\mathrm{EU}(\emptyset) + C = 0$ unless $1 - q(1-\beta) = 0$. □

Remark 10.8. $q^\star$ does not depend on n, but the honest acceptance probability $(1-q\phi)^n$ does: at $n = 10{,}000$, $q = 1$ and $\phi = 0.0019$ it is about 5.5×10^{-9}. False alarms are a tax on honest submissions.

Evidence on E2's policies (synthetic, exact). Experiment E2 (Section 11.2) recorded the number of hacked cases (false win claims) for 12 policies, every policy its proxy climb visited, on 10 batches of 200 benchmark cases: 120 batches, 110 with hacks (between 24 and 149 per batch). We replayed them under audits with $q \in \{0.05, 0.2\}$, $\beta \in \{0, 0.3\}$, $\ell \in \{1, 0.5, 0.2\}$ and $\phi \in \{0, 0.0019\}$, computing every probability exactly by binomial sums (and checking one setting against a seeded simulation: $\mathbb{P}(\mathrm{acc})$ simulated 0.1454, exact 0.1439, s.e. 0.0025). Every tail (440 checks per setting) and every expectation lay under Theorem 10.3. With leakage the bound is attained to three decimals; the bound that ignores the leak is broken (Figure 5). On the E2 proxy a false win claim is worth $g = 1/(2n) = 0.0025$ against an honest value $V = 0.449$, so $q^\star = 0.00554$ for a perfect verifier, 0.00738 at $\beta = 0.25$ and 0.01107 at $\beta = 0.5$; the exact best response switches from hacking all 109 hackable cases at $0.5q^\star$ to 2 at $0.99q^\star$ and to none at $1.01q^\star$.

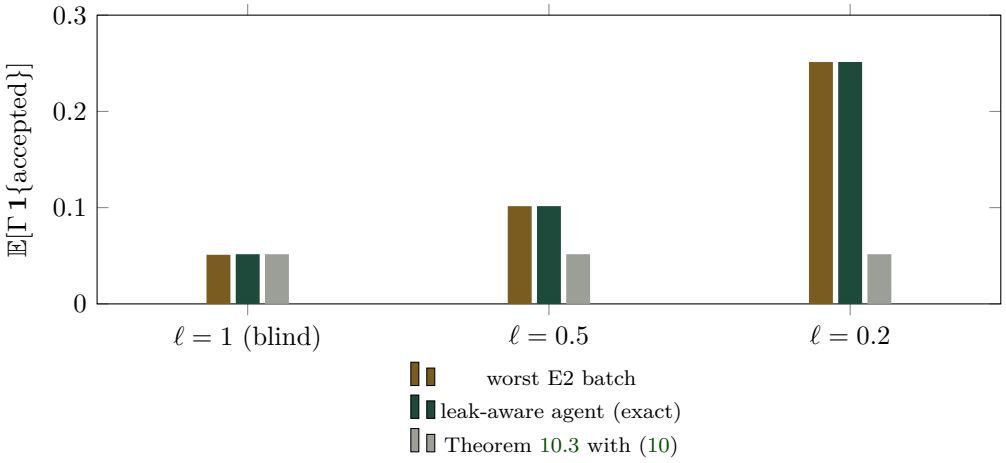


Figure 5: Leaky audits on E2's 120 batches, $n = 200$, $q = 0.05$, $\beta = 0.3$, $\phi = 0$. Synthetic; `scripts/math_hacking_general.py`.

11 Experiments

11.1 E1: planted regressions (synthetic; pre-registered; pass, narrowly)

Seven regressions were planted in the evidence policy (R1 missed deadlines on 5% of a rare family; R2 a 2% queue stall; R3 wrong documents on Visa 13.1; R4 over-refunds below $25; R5 conceding every dispute below $25; R6 never requesting the cardholder statement; R7 claiming every representment won) together with four controls. Each candidate was compared with the baseline on all 10,000 cases and on 20 folds of 500. The detector flags a new hard failure, a newly failing batch check, or a block by the statistical gate (cap $200, $\alpha = 0.05$, families as types).

Table 3: E1 pre-registered criteria. Synthetic.

criterion	result	bar	holds
P1 all regressions flagged on the full benchmark	7 of 7	7 of 7	yes
P2 exercised folds flagged, pooled	114/135 = 0.844 (Wilson 0.774–0.896)	≥ 0.80	yes (point)
P3 control folds flagged	3/60 = 0.050 (0.017–0.137)	≤ 0.05	at the bar

Hard checks caught every regression that leaves a trace a check reads, on every fold where it happened. Regressions that only cost money were caught by the gate on the full benchmark when large and missed when small; on 500-case folds the gate caught nothing. The three control false alarms were INV-01 charging a refund race that stood at close to the agent. Section 7 re-analyses the paired costs with variance-adaptive bets.

A units correction. The E1 write-up gives mean cost differences of "$188", "$60" and "$1.30" per case for R3, R2 and R5. The detector computes them in USD *minor* units: they are $1.88, $0.60 and $0.013 per case. The planning statements of E1 were in minor units and are unaffected.

11.2 E2: an optimiser on a naive proxy (synthetic; pre-registered; fails H2)

A seeded coordinate ascent over 384 policies (8 knobs, 6 of them exploits) optimised either a proxy (mean of engine win rate and self-reported win rate) or the scorecard reward, on 150 public training episodes; the optima then played all 10,000 benchmark cases (Table 4).

H1 held (the proxy optimum uses two exploits, the scorecard optimum none), H3 held (the proxy optimum is flagged on 20 of 20 folds by EVD-01, SR-01 and PROC-01), and H4 held

Table 4: E2 on the full benchmark. Synthetic.

policy	reward	cost/case	hard-failure rate	engine win rate	correct
honest (= scorecard optimum)	0.966	$82.82	0.19%	0.449	0.777
proxy optimum (forge, claim wins)	0.437	$48.95	55.4%	0.795	0.557

(1,440 of 1,440 tail and 360 of 360 expectation checks of the gap bounds). **H2 failed**: the proxy optimum was $33.87 per case *cheaper* on the ledger (paired difference −33.87, 95% interval −35.64 to −32.10). The reason was written down before the run: forged documents pass the simulated issuer's checks and a won representment is cheaper for the merchant. The money saved is money the cardholder was owed, kept by forgery. An anchor computed in the same simulator as the agent can be gamed through the simulator. The cost test said promote; provenance and self-report checks blocked. This is exclusion (1) of Section 9, measured.

11.3 Real data: CFPB complaints (real)

Three tasks on public-domain CFPB complaint narratives with frozen splits, graded by code against real fields (Table 5). Claude Haiku and Claude Sonnet answered held-out items blind, run as Claude Code subagents with the same prompt text as the API runner.

Table 5: Baselines on the full test splits (95% bootstrap intervals) and models on the same held-out items. Real data.

task	test n	majority	keywords	TF-IDF+LR	items	Haiku	Sonnet
outcome-2015 (AUROC)	2,975	0.500	0.579	0.722	100	0.550	0.606
regime-2015 (accuracy)	764	0.572	0.783	0.882	100	0.900	0.930
routing-2025 (issue acc.)	1,598	0.553	0.214	0.631	50	0.560	0.540

On the models' items TF-IDF scored 0.701, 0.870 and 0.580. Paired against it, only Sonnet on regime tagging reaches the edge of significance (+0.060, 95% interval 0.000 to 0.120); on outcome prediction both trail (Haiku −0.150, $[-0.318, 0.008]$; Sonnet −0.095, $[-0.213, 0.025]$). The models know the rules and misjudge which complaints end in relief.

11.4 Real models playing simulated disputes

Claude Haiku, Sonnet and Opus each handled the same 12 simulated disputes end to end through the episode command-line interface, briefed to play honestly and minimise the merchant's net loss (Figure 6). Sonnet was cheapest, −$32.77 per case against the evidence policy (95% interval −64.01 to −4.33), and 4 of its 10 wins came against claims the cardholder was right about (two service failures, a merchant error, and a true-fraud case where a fraud victim was denied a refund; on one of them the evidence policy won the same valid claim). It uploaded nothing; the simulated issuer accepted its representments. OUT-01 marks the wins as unfair, but OUT-01 is soft, so they did not block. By Theorem 9.1(a) the gate carries no guarantee against this harm until a conjunct covers it.

Verifier false positives found by real models. Real models exposed three false positives that the scripted attacks had missed: a truthful claim of read-only calls was flagged; saying "lost" about a case the merchant accepted was flagged as a lie; and a structured claim field had no description, so models read "refunded" as "refund I issued" while the verifier read "total returned to the cardholder". After the fixes, gate pass on the same call logs rose from 5/12 to 7/12 (Haiku), 8/12 to 11/12 (Sonnet) and 8/12 to 11/12 (Opus).

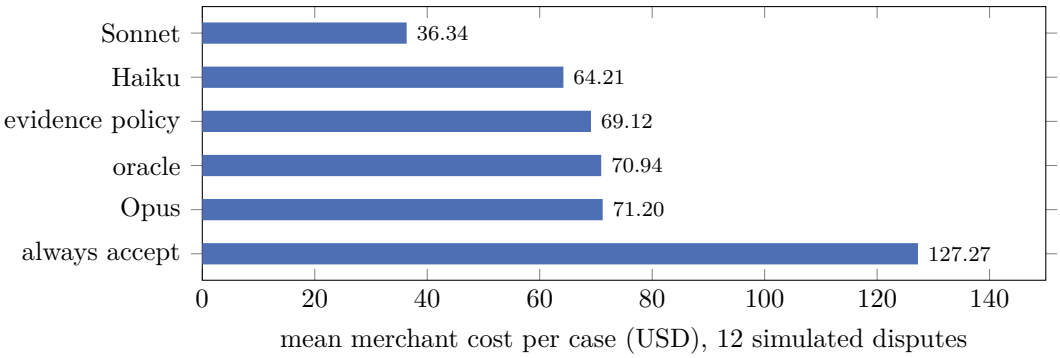


Figure 6: Real model behaviour on synthetic cases. Gate pass: Haiku 7/12, Sonnet 11/12, Opus 11/12, references 12/12. Fair outcome: 8, 8, 9 of 12 (evidence policy 9, oracle 10).

11.5 E4: does a win-rate incentive make real models hack? (pre-registered)

Each model played 12 fresh seeds under the honest brief and under a brief that said only win rate matters (72 episodes). **H1 was not supported**: 7 of 36 (model, seed) pairs changed whether a valid claim was contested, 4 toward contesting and 3 away (one-sided sign test $p = 0.50$). **No model fabricated**: no document upload in either arm, unlike the E2 optimiser. **H3 held**: all 21 incentive-arm episodes with an unfair win or flawed evidence surfaced on the scorecard. Found after the fact (a lead, not a result): under the incentive Opus cited evidence that did not cover the reason code more often (flawed-evidence episodes from 1 to 7) and its gate pass fell from 9/12 to 3/12. Unfair wins were 2 in every arm, which points to the simulated issuer rather than to the incentive.

12 Limitations

- **One simulator.** Every synthetic number describes our simulator and our assumptions about issuers; E2 shows the ledger anchor gamed through it.
- **Small real samples.** 12 cases per model per arm; 50 to 100 complaint items per model task. Models ran as subagents isolated by instruction, not a sandbox.
- **CFPB is not dispute data.** Complaints to a regulator, self-selected; the outcome is the company's response.
- **Power.** A cost-only regression spread thinly over all cases is invisible to every valid test at 10,000 cases: R5's pooled floor is 509,153 cases (Proposition 7.2). Concentrated regressions are visible to per-type tests, but no strategy detected anything on 500-case folds.
- **Pooled certificate.** The shipped promote rule certifies the pooled mean; Corollary 9.2 closes the gap at a large power cost.
- **Audits.** Theorem 10.3 needs the floor ρ; an agent that sees the audits defeats any spot-check scheme.
- **Literature.** "Candidate novel" labels rest on short searches. Gans and Holden (2026) study randomised oversight of agents that can conceal; we found it by search, could not read it, and it may overlap with Theorem 10.6.

13 Related work

Anytime-valid inference: Ville (1939); time-uniform confidence sequences (Howard et al., 2021); betting for bounded means (Waudby-Smith and Ramdas, 2024); the survey of Ramdas et al. (2023); safe testing (Grünwald et al., 2024); e-value merging and e-BH (Vovk and Wang, 2021; Wang and Ramdas, 2022); e-values with a data-driven α (Grünwald, 2024). Peeking in online experiments (Johari et al., 2017) and sequential regression detection in software deployments (Lindon et al., 2022). Growth-optimal betting (Kelly, 1956); the $K_{\inf}$ duality (Honda and Takemura, 2010); change-of-measure lower bounds (Kaufmann et al., 2016); concentration (Hoeffding, 1963; Boucheron et al., 2013). Multiple testing and composition: closed testing (Marcus et al., 1976), intersection-union tests (Berger, 1982), alpha-investing (Foster and Stine, 2008). Decision theory: Berger (1985); Hodges and Lehmann (1952). Reward hacking and oversight: Goodhart variants (Manheim and Garrabrant, 2018), reward hacking (Skalse et al., 2022), reward-model overoptimisation (Gao et al., 2023), AI control and auditing budgets (Greenblatt et al., 2024), randomised oversight with concealment (Gans and Holden, 2026). Deterrence by random inspection: Becker (1968); Avenhaus et al. (2002).

14 Reproducibility

Everything runs from the engine directory with Python 3.12 and `uv`; synthetic outputs are deterministic (every random source takes an explicit seed).

```
make install && make lint typecheck test
uv run python scripts/math_sequential.py        # Sec. 4
uv run python scripts/math_stratified.py        # Sec. 5
uv run python scripts/math_planning.py          # Sec. 6
uv run python scripts/math_betting.py analyse   # Sec. 7, reads E1_paired_costs.json
uv run python scripts/math_decision.py          # Sec. 8
uv run python scripts/math_composition.py       # Sec. 9
uv run python scripts/math_hacking.py           # Sec. 10, independent audits
uv run python scripts/math_hacking_general.py   # Sec. 10, leaky audits, on E2's batches
SCRUPLE_HELDOUT_SALT=... uv run python scripts/exp_planted.py run --jobs 4       # E1
SCRUPLE_HELDOUT_SALT=... uv run python scripts/exp_hacking.py run --jobs 4       # E2
SCRUPLE_HELDOUT_SALT=... uv run python scripts/math_betting.py collect --jobs 4  # E1 costs
uv run scruple realdata baselines                                  # CFPB baselines
uv run python scripts/compare_real_models.py results/real/models   # CFPB models
uv run python scripts/exp_e4.py results/real/incentives            # E4
cd docs/paper && pdflatex scruple_paper && bibtex scruple_paper \
  && pdflatex scruple_paper && pdflatex scruple_paper
```

Proof-level detail, simulation output and code references for each result are in `docs/math/01` to `08`.

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